

Homework 6: Metals

Paper track. No home lab (the metals kit stays at practice).

Calculator needed.

Part A: Identify the metal (2 pts each)

#	Color	Magnet	Density (g/cm ³)	With HCl	Metal + symbol + atomic #
1	reddish-orange	no	8.96	no reaction	
2	silvery	no	2.70	slow start, then steady	
3	gray	yes	7.87	slow-moderate, solution goes pale green	
4	dull silver	no	1.74	vigorous, fast	
5	bluish-gray	no	7.14	moderate, steady	

Part B: Density (3 pts each)

1. A metal cube 3.0 cm on a side weighs 72.9 g. Density? Which metal?
2. A piece of metal raises water in a graduated cylinder from 25.0 mL to 34.0 mL and weighs 80.6 g. Density? Which metal?
3. Convert 7.14 g/cm³ to kg/m³.
4. A solid iron bar has a volume of 50 cm³. What is its mass? (Fe = 7.87 g/cm³)
5. A 2.0-meter cube of aluminum — what is its mass in kg? (Al = 2700 kg/m³)

Part C: Reactions & oxides (2 pts each)

1. Classify: $2 \text{H}_2\text{O}_2 \rightarrow 2 \text{H}_2\text{O} + \text{O}_2$
2. Classify: $\text{Zn} + 2 \text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$
3. Classify: $2 \text{Mg} + \text{O}_2 \rightarrow 2 \text{MgO}$

4. Why does aluminum resist acid at first, but iron just keeps corroding?
5. Steel, brass, and bronze — name the metals in each.

Part D: Scenario (3 pts)

A gray metal fragment at a scene is not attracted to a magnet, has a measured density of 2.71 g/cm^3 , and reacts slowly with HCl at first and then steadily. Name the metal. A suspect who works with aircraft parts and a suspect who works with cast-iron cookware are both of interest — which does this implicate?

Answer Key

Part A: 1. Copper — Cu — 29. 2. Aluminum — Al — 13. 3. Iron — Fe — 26. 4. Magnesium — Mg — 12. 5. Zinc — Zn — 30.

Part B:

1. Volume = 27.0 cm^3 ; $72.9 \div 27.0 = \mathbf{2.70 \text{ g/cm}^3} \rightarrow \mathbf{\text{aluminum}}$.
2. Volume = 9.0 mL ; $80.6 \div 9.0 \approx \mathbf{8.96 \text{ g/cm}^3} \rightarrow \mathbf{\text{copper}}$.
3. $\mathbf{7140 \text{ kg/m}^3}$.
4. $7.87 \times 50 = \mathbf{393.5 \text{ g}}$.
5. $(2.0)^3 = 8.0 \text{ m}^3$; $8.0 \times 2700 = \mathbf{21,600 \text{ kg}}$.

Part C:

1. Decomposition. 2. Single replacement. 3. Synthesis (combination).
2. Aluminum forms a thin, tough, **protective** oxide layer (Al_2O_3) that seals the surface (passivation); iron's oxide (rust, $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$) is flaky and does not protect, so oxygen keeps reaching fresh metal.
3. Steel = iron + carbon; brass = copper + zinc; bronze = copper + tin.

Part D: Aluminum (non-magnetic, density ~ 2.70 , oxide-layer delay then steady reaction). It implicates the **aircraft-parts** suspect (aircraft use aluminum alloys); the cast-iron-cookware suspect is cleared — that would be iron, which is magnetic and much denser.